

- 7 (a) The fruit fly, *Drosophila melanogaster*, usually has red eyes. A gene for eye colour has four alleles: red, apricot, honey and white.

Define the terms *gene* and *allele*.

gene

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allele

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[2]

- (b)
- The allele for red eyes, C^R , is dominant to the other three alleles.
 - The allele for apricot eyes, C^A , is dominant to the allele for honey eyes, C^H .
 - The allele for white eyes, C^W , is recessive to the other three alleles.

Construct a genetic diagram to show the genotypes, phenotypes and ratio of the offspring from a cross between a fruit fly with red eyes, $C^R C^H$, and a fruit fly with apricot eyes, $C^A C^W$.

parents phenotypes

red eyes

apricot eyes

parents genotypes

$C^R C^H$

$C^A C^W$

gametes

offspring genotypes

offspring phenotypes

ratio

[3]

- (c) Describe how you would carry out a test cross **and** use it to determine the genotype of a red-eyed fruit fly.

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[Total: 8]